

[20 marks]

Exercise I.

Compute the following integrals:

1. $\int \frac{\cos x}{4 + \sin^2 x} dx,$

2. $\int \frac{\cos x \ln(\sin x)}{\sin x} dx.$

3. $\int \frac{4x + 1}{x(2x + 1)} dx,$

4. $\int \frac{-2x^3 - x^2 + 4x + 1}{2x^2 + x} dx.$

[10 marks]

Exercise II.

Let $U_n = \sum_{k=n}^{2n} \frac{1}{k}$

1. Show that $U_n = \frac{1}{n} + \sum_{k=1}^n \frac{1}{k+n}$

2. Calculate $\lim_{n \rightarrow \infty} U_n$

[25 marks]

Exercise III.

Study the natures of the following improper integrals :

1. $\int_0^{\infty} \frac{\sin t}{1+t^2} dt,$

2. $\int_0^{\infty} \sin(t^2 + 1) dt.$

3. $\int_1^{\infty} \frac{t+1}{\sqrt{t-1}(t^2+1)} dt,$

4. $\int_0^{\frac{\pi}{2}} (\tan t)^{\frac{3}{2}} dt.$

[20 marks]

Exercise IV.

Study the natures of the following series:

1. $\sum_{n=1}^{\infty} \frac{\ln n}{n},$

2. $\sum_{n=1}^{\infty} \frac{(-1)^n + n}{n^2 + n + 1},$

3. $\sum_{n=1}^{\infty} n^{\frac{1}{n}},$

4. $\sum_{n=1}^{\infty} \frac{\arctan n}{\sqrt{n} + n^2}.$

[25 marks]

Exercise V.

Let $I = \int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx$ where $0 < a < b$ and $f : [0, \infty) \mapsto \mathbb{R}$ be a **continuous** function with $\lim_{x \rightarrow \infty} f(x) = \ell \in \mathbb{R}$.

1. Show that for $X > \varepsilon > 0$ we have

$$\int_{\varepsilon}^X \frac{f(ax) - f(bx)}{x} dx = \int_{\varepsilon a}^{\varepsilon b} \frac{f(t)}{t} dt - \int_{Xa}^{Xb} \frac{f(t)}{t} dt.$$

2. Using the first mean value theorem, show that

$$\lim_{\varepsilon \rightarrow 0} \int_{\varepsilon a}^{\varepsilon b} \frac{f(t)}{t} dt = f(0) \ln \frac{b}{a} \text{ and } \lim_{X \rightarrow \infty} \int_{Xa}^{Xb} \frac{f(t)}{t} dt = \ell \ln \frac{b}{a}.$$

3. Deduce that I converges and give its value.



Exercice I. Compute the following integrals:

1. $\int \frac{\cos x}{4 + \sin^2 x} dx \xrightarrow{\sin x = t} \int \frac{dt}{4 + t^2} = \frac{1}{2} \arctan(t/2) + C = \frac{1}{2} \arctan\left(\frac{\sin x}{2}\right) + C.$

2. $\int \frac{\cos x \ln(\sin x)}{\sin x} dx \xrightarrow{\sin x = t} \int \frac{\ln t}{t} dt \xrightarrow{t = u} \int \frac{1}{u} du = \ln|u| + C = \ln\left(\frac{1}{2}\right) + C = \ln\left(\frac{1}{2}\right) + C.$

or $\int \frac{\cos x \ln(\sin x)}{\sin x} dx \xrightarrow{\ln(\sin x) = t} \int t dt = \frac{t^2}{2} + C = \frac{(\ln(\sin x))^2}{2} + C.$

3. $\int \frac{4x+1}{x(2x+1)} dx = \int \frac{1}{x} dx + \int \frac{2}{2x+1} dx = \ln|x| + \ln|2x+1| + C$

Decomposition: $\frac{4x+1}{x(2x+1)} = \frac{A}{x} + \frac{B}{2x+1}$

$\Rightarrow 4x+1 = A(2x+1) + Bx$

$x=0 \Rightarrow A=1$ $x=1 \Rightarrow 5=3+B \Rightarrow B=2$

4. $\int \frac{-2x^3 - x^2 + 4x + 1}{2x^2 + x} dx = \int \left(-x + \frac{4x+1}{2x^2+x} \right) dx$

Eucl. Divis: $\begin{array}{r} 2x^2+x \overline{) -2x^3-x^2+4x+1} \\ \underline{-2x^2-x} \\ 4x+1 \end{array}$

$\frac{4x+1}{2x^2+x} = \frac{1/2}{x} + \ln|2x+1| + C$

Exercice II

Let $u_n = \sum_{k=n}^{\infty} \frac{1}{k}$

1. S.T. $u_n = \frac{1}{n} + \sum_{k=1}^{\infty} \frac{1}{k+n}$

$u_n = \sum_{k=n}^{\infty} \frac{1}{k} = \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n} = \frac{1}{n} + \sum_{k=1}^n \frac{1}{k+n}$

2. calculate $\lim_{n \rightarrow \infty} u_n$

$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k+n} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n(k_n+1)} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{k_n+1}$

$\delta(n) = \frac{1}{n+1}$ is cont, hence Riemann on $[0,1]$.

$= \int_0^1 \frac{1}{1+t} dt = \ln|1+t| \Big|_0^1 = \ln 2 - \ln 1 = \ln 2.$

Exercice III. 1. $I_1 = \int_0^{\infty} \frac{\sin t}{1+t^2} dt$; $\left| \frac{\sin t}{1+t^2} \right| \leq \frac{1}{1+t^2} \leq \frac{1}{t^2}$. Since $\int_1^{\infty} \frac{1}{t^2} dt$ conv, then by comp test I_1 is absol conv.

2. $I_2 = \int_0^{\infty} \sin(t^{1/2}) dt = \int_0^{\infty} \frac{2t \sin(t^{1/2})}{2t^{1/2}} dt$; $\lim_{t \rightarrow \infty} \frac{2t \sin(t^{1/2})}{2t^{1/2}} = 0$ & $F(x) = \int_0^x \sin(t^{1/2}) dt = -\cos(t^{1/2}) \Big|_0^x$

$\Rightarrow |F(x)| = |\cos(x^{1/2}) - \cos(0)| \leq 2$ on $[0, \infty)$

$\therefore I_2$ conv by Abel-Dire. [2nd way integ by parts or Cauchy Criter]

3. $I_3 = \int_1^{\infty} \frac{t+1}{\sqrt{t-1}(t^2+1)} dt$; in $N(2)$: $f(t) = \frac{2}{2(t-1)^{1/2}}$ Hence $\int_1^2 f(t) dt$ conv by Riemann

4. $\int_0^{\infty} (\tan t)^{3/2} dt$ $t = \arctan u$ $\int_0^{\infty} \frac{u^{3/2}}{1+u^2} du$ div by Riemann $\frac{u^{3/2}}{1+u^2} \sim \frac{1}{u^{1/2}}$

Exercise 4. Study the nature of the following series:

⑤ 1. $\sum_{n=1}^{\infty} \frac{\ln n}{n} = \sum_{n=1}^{\infty} \frac{1}{n^1 (\ln n)^1}$ Root-test div ($p=1$ & $q=-1 \leq 1$)

2. $\sum_{n=1}^{\infty} \frac{(-1)^n + n}{n^2 + n + 1}$; $\frac{(-1)^n + n}{n^2 + n + 1} \sim \frac{n}{n^2} = \frac{1}{n} > 0$. The series div (By equiv test & known $\sum \frac{1}{n}$ div)

or $\sum \frac{(-1)^n}{n^2 + n + 1}$ conv (Alt ser) & $\sum \frac{n}{n^2 + n + 1}$ div (equiv test). Thus the series div as sum of conv & div series

or $\frac{(-1)^n + n}{n^2 + n + 1} > \frac{1}{n}$ for n large. $\sum \frac{1}{n}$ div (equiv test) $\Rightarrow$ Series div by Comparison

3. $\sum_{n=1}^{\infty} n^{1/n} = \sum_{n=1}^{\infty} e^{\frac{1}{n} \ln n}$; $\lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln n} = e^0 = 1 \neq 0 \Rightarrow$ The series diverges

4. $\sum_{n=1}^{\infty} \frac{\arctan n}{\sqrt{n} + n^2}$; $\frac{\arctan n}{\sqrt{n} + n^2} \sim \frac{\pi/4}{n^2} > 0$. $\sum \frac{1}{n^2}$ conv $\Rightarrow$ The series conv. (Ratio test $d=2$)

or $n^2 \cdot \frac{\arctan n}{\sqrt{n} + n^2} \rightarrow \pi/4 \neq 0$ & $d=2 > 1$. Then the series conv absol

or $n^{3/2} \cdot \frac{\arctan n}{\sqrt{n} + n^2} \rightarrow 0$ & $d=3/2 > 1$. Then the series conv absol.

Exercise 5. $I = \int_0^x \frac{f(ax) - f(bx)}{x} dx$, $0 < a < b$, $f: (0, \infty) \rightarrow \mathbb{R}$ cont & $\lim_{x \rightarrow 0} f(x) = l \in \mathbb{R}$.

1. S.T for $x > \varepsilon > 0$; $\int_{\varepsilon}^x \frac{f(ax) - f(bx)}{x} dx = \int_{\varepsilon a}^x \frac{f(t)}{t} dt - \int_{\varepsilon b}^x \frac{f(t)}{t} dt$.

$\int_{\varepsilon}^x \frac{f(ax)}{x} dx - \int_{\varepsilon}^x \frac{f(bx)}{x} dx = \int_{\varepsilon a}^{ax} \frac{f(t)}{t} dt - \int_{\varepsilon b}^{bx} \frac{f(t)}{t} dt$; $t = ax$ and $s = bx$
 $= \int_{\varepsilon a}^{ax} \frac{f(t)}{t} dt + \int_{\varepsilon b}^{bx} \frac{f(t)}{t} dt - \left[\int_{\varepsilon a}^{ax} \frac{f(s)}{s} ds + \int_{\varepsilon b}^{bx} \frac{f(s)}{s} ds \right] = \int_{\varepsilon a}^{ax} \frac{f(t)}{t} dt - \int_{\varepsilon b}^{bx} \frac{f(t)}{t} dt$

2. Using P.T.M.V.T. S.T $\lim_{\varepsilon \rightarrow 0} \int_{\varepsilon a}^{ax} \frac{f(t)}{t} dt = f(a) \ln \frac{b}{a}$ & $\lim_{x \rightarrow 0} \int_{\varepsilon b}^{bx} \frac{f(t)}{t} dt = l \ln \frac{b}{a}$

~~$\lim_{x \rightarrow 0} \int_{\varepsilon a}^{ax} \frac{f(t)}{t} dt = f(a) \ln \frac{b}{a}$~~

$\int_{\varepsilon a}^{ax} \frac{f(t)}{t} dt \stackrel{P.T.M.V.T.}{=} f(c_{\varepsilon}) \int_{\varepsilon a}^{ax} \frac{1}{t} dt = f(c_{\varepsilon}) [\ln \varepsilon b - \ln \varepsilon a] = f(c_{\varepsilon}) \ln \frac{b}{a}$ $\begin{cases} f \in C([a, b]) \\ g(t) = 1/t > 0 \\ \varepsilon a < c_{\varepsilon} < \varepsilon b \end{cases}$

$\lim_{\varepsilon \rightarrow 0} \int_{\varepsilon a}^{ax} \frac{f(t)}{t} dt = \lim_{\varepsilon \rightarrow 0} f(c_{\varepsilon}) \ln \frac{b}{a} = f(0) \ln \frac{b}{a}$ [$\because f$ cont at 0]

$\int_{\varepsilon b}^{bx} \frac{f(t)}{t} dt \stackrel{P.T.M.V.T.}{=} f(c_x) \int_{\varepsilon b}^{bx} \frac{1}{t} dt = f(c_x) [\ln bx - \ln \varepsilon b] = f(c_x) \ln \frac{x}{\varepsilon}$ $\begin{cases} f \in C([a, b]) \\ g(t) = 1/t > 0 \\ \varepsilon b < c_x < bx \end{cases}$

$\lim_{x \rightarrow 0} \int_{\varepsilon b}^{bx} \frac{f(t)}{t} dt = \lim_{x \rightarrow 0} f(c_x) \ln \frac{x}{\varepsilon} = l \ln \frac{b}{a}$

3. Deduce I converges & give its value
 $I = \lim_{\varepsilon \rightarrow 0} \lim_{x \rightarrow 0} \int_{\varepsilon}^x \frac{f(ax) - f(bx)}{x} dx \stackrel{1.}{=} \lim_{\varepsilon \rightarrow 0} \lim_{x \rightarrow 0} \left[\int_{\varepsilon a}^{ax} \frac{f(t)}{t} dt - \int_{\varepsilon b}^{bx} \frac{f(t)}{t} dt \right] \stackrel{2.}{=} f(0) \ln \frac{b}{a} - l \ln \frac{b}{a}$
 Thus I converges.